

A

Real-time research project on

**MoodChef : Mood-Based Smart Recipe Recommender
with Cooking Assistance**

Submitted in partial fulfillment of the requirements for the award of degree

**BACHELOR OF TECHNOLOGY IN
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)**

Submitted By

S.SAI DEEPTHI (23TQ1A6694)

SHAIK NOOR ALI (23TQ1A6688)

K.DILEEP KUMAR (23TQ1A66A7)

Under the Guidance of

Dr. A.SATYA NARAYANA

Assistant Professor



SIDDHARTHA

INSTITUTE OF TECHNOLOGY & SCIENCE

(UGC AUTONOMOUS)

**(Affiliated to JNTUH, Approved by AICTE, Accredited by NBA & NAAC with A Grade, nirf
Ranked & An ISO Certified Institution)**

Narapally (V), Ghatkesar (M), Medchal (D), TS-500088

2024-2025



SIDDHARTHA

INSTITUTE OF TECHNOLOGY AND SCIENCE

UGC - AUTONOMOUS

(Affiliated to JNTUH, Approved by AICTE, Accredited by NBA* & NAAC
with A+ Grade, nirf Ranked & An ISO Certified Institution)

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING -ARTIFICIAL
INTELLIGENCE & MACHINE LEARNING**

CERTIFICATE

This is to certify that the project report entitled MoodChef:MOOD-BASED SMART RECIPE
RECOMMENDER WITH COOKING ASSISTANCE being submitted by

S.SAI DEEPTHI (23TQ1A6694)

SHAIK NOOR ALI (23TQ1A6688)

K.DILEEP KUMAR (23TQ1A66A7)

In partial fulfilment for the award of the degree of Bachelor of Technology in Computer Science
and Engineering(AI&ML), Siddhartha Institute Of Technology And Sciences UGC-
Autonomous, is a record of Bonafide work carried out under my guidance and supervision. The
results embodied in this project report have not been submitted to any other University or
Institute for the award of any Degree or Diploma.

Guide

Dr. A. SATYA NARAYANA M.Tech, PhD

Head of the Department

TSKS. JYOTHIRMAYI M.Tech,(PhD)

Internal Examiner

External Examiner

PRINCIPAL

DECLARATION

We declare that this project report titled MoodChef: MOOD-BASED SMART RECIPE RECOMMENDER WITH COOKING ASSISTANCE submitted in partial fulfilment of the degree of B. Tech in CSE(AI&ML) is a record of original work carried out by us under the supervision of MR.A. SATYANARAYANA and has not formed the basis for the award of any other degree or diploma, in this or any other Institute or University. In keeping with the ethical practice in reporting scientific information, due acknowledgments have been made wherever the findings of others have been cited.

DATE:

SIGNATURE:

S.SAI DEEPTHI	(23TQ1A6694)
SHAIK NOOR ALI	(23TQ1A6688)
K.DILEEP KUMAR	(23TQ1A66A7)

ACKNOWLEDGEMENT

We express our sincere gratitude to beloved and highly esteemed institute SIDDHARTHA INSTITUTE OF TECHNOLOGY AND SCIENCES for grooming us into Computer Science and Engineering graduate, we wish to thank PRINCIPAL DR. M. JANARDHAN for Providing a great learning environment.

We profoundly express our sincere thanks to JYOTHIRMAYI, Head Department of ARTIFICIAL INTELLIGENCE & MACHINE LEARNING, SITS, for her cooperation and encouragement in completing the project successfully.

We wish to record our deep sense of gratitude to our Project In-charge Dr. A. Satya Narayana, Assistant Professor, in ARTIFICIAL INTELLIGENCE & MACHINE LEARNING, SITS, for giving his insight, advice provided during the review sessions and providing constant monitoring from time to time in completing our project and for giving us this opportunity to present the project work.

We express our sincere gratitude to our guide Dr. A. Satya Narayana, Assistant Professor, in ARTIFICIAL INTELLIGENCE & MACHINE LEARNING Department, SITS, who motivated throughout the period of the project and also for his valuable and intellectual suggestions apart from his adequate guidance, constant encouragement right throughout our work.

Finally, we would like to thank Project Co-Ordinator, Project Review Committee (PRC)members, all the faculty members and supporting staff, Department of Computer Science and Engineering, SITS for extending their help in all circumstance.

S.SAI DEEPTHI	(23TQ1A6694)
SHAIK NOOR ALI	(23TQ1A6688)
K.DILEEP KUMAR	(23TQ1A66A7)

TABLE OF CONTENTS

DESCRIPTION	PAGE NUMBER
CERTIFICATE	II
DECLARATION	III
ACKNOWLEDGEMENT	IV
ABSTRACT	1
CHAPTER 1: INTRODUCTION	2
1.1 PROBLEM STATEMENT	3
1.2 OBJECTIVE OF PROJECT	4
1.3 SCOPE OF THE PROJECT	4
1.4 MOTIVATION	5
CHAPTER 2: LITERATURE SURVEY	6
CHAPTER 3: EXISITING SYSTEM	10
3.1 DRAWBACKS OF EXISTING SYSTEM	12
CHAPTER 4: PROPOSED SYSTEM	13
CHAPTER 5: SOFTWARE AND HARDWARE REQUIREMENTS	15
5.1 SOFTWARE REQUIREMENTS	16
5.2 HARDWARE REQUIREMENTS	16
CHAPTER 6: SYSTEM DESIGN	17
6.1 SOFTWARE DESIGN	18
6.2 UML DIAGRAM	19
6.2.1 USE CASE DIAGRAM	19

6.2.2 CLASS DIAGRAM	20
6.2.3 SEQUENCE DIAGRAM	21
6.2.4 ACTIVITY DIAGRAM	22
6.3 MODULES	22
CHAPTER 7: IMPLEMENTATION	25
7.1 SAMPLE SOURCE CODE	28
CHAPTER 8: RESULTS	34
CHAPTER 9: CONCLUSION	40
CHAPTER 10: FUTURE SCOPE	41
REFERENCES	43

LIST OF FIGURES

S.NO	FIGURE NO.	NAME	PAGE NO.
1	Fig 6.1	Architectural Design	16
2	Fig 6.2.1	Use Case Diagram	17
3	Fig 6.2.2	Class Diagram	18
4	Fig 6.2.3	Sequence Diagram	19
5	Fig 6.2.4	Activity Diagram	19
6	Fig 8.1	Output snapshot 1	35
7	Fig 8.2	Output snapshot 2	35
8	Fig 8.3	Output snapshot 3	34
9	Fig 8.4	Output snapshot 4	34
10	Fig 8.5	Output snapshot 5	37
11	Fig 8.6	Output snapshot 6	37
12	Fig 8.7	Output snapshot 7	38
13	Fig 8.8	Output snapshot 8	38

ABSTRACT

This project introduces **MoodChef: Mood-Based Smart Recipe Recommender With Cooking Assistant**. The system personalizes recipe suggestions based on the user's **emotional state, dietary preferences, and nutritional goals** using modern **AI techniques** such as **Natural Language Processing (NLP)** and **facial emotion recognition**. By leveraging **face-api.js** for real-time mood detection and implementing **smart filtering algorithms**, **Mood-Based Smart Recipe Recommender With Cooking Assistant** identifies the most suitable recipes from a curated dataset.

The platform also features a **voice-enabled assistant** that provides intuitive, step-by-step **cooking guidance**, enhancing accessibility and enabling a hands-free user experience. Designed to make cooking more **enjoyable, personalized, and health-conscious**, **Mood-Based Smart Recipe Recommender With Cooking Assistant** addresses common barriers like decision fatigue and lack of motivation in the kitchen.

Additional functionalities include an adaptive user interface, seamless preference selection, and support for vegan, vegetarian, and non-vegetarian categories. The system architecture ensures that even novice users can comfortably navigate and cook healthy meals tailored to their mood.

CHAPTER 1

INTRODUCTION

INTRODUCTION

In recent years, the fusion of artificial intelligence (AI) with human-centric applications has given rise to innovative solutions that prioritize personalization, convenience, and emotional intelligence. One such domain where AI can significantly enhance user experience is culinary assistance and recipe recommendation. Cooking, while essential, can often feel like a repetitive or overwhelming task, especially when users are unsure of what to prepare or lack motivation. Emotional state plays a critical role in influencing food choices, appetite, and even cooking engagement. Despite the growing number of recipe and meal-planning apps available, very few incorporate emotional context into their recommendation logic.

To address this gap, we present a mood-based smart recipe recommender with cooking assistance that dynamically responds to the user's emotional state to suggest personalized recipes. The application leverages facial expression recognition technology to analyze users' real-time emotions using their webcam. This input is then combined with user-provided dietary preferences (such as vegan, vegetarian, or non-vegetarian) to filter and recommend appropriate recipes that not only align with the user's nutritional needs but also resonate with their emotional well-being.

What sets MoodChef:Mood-Based Smart Recipe Recommender With Cooking Assistance apart is its voice-enabled cooking assistant, which provides step-by-step spoken guidance to users during meal preparation. This hands-free functionality makes it especially useful in kitchen environments where visual or tactile interaction with devices may be impractical. The application is built using web technologies such as HTML, CSS (via Tailwind CSS), and JavaScript, with face-api.js used for facial expression detection and the Web Speech API for voice output.

1.1 PROBLEM STATEMENT

- In today's fast-paced lifestyle, individuals often struggle to decide what to cook or eat, especially when their emotional state affects motivation and appetite. Traditional recipe platforms fail to consider the **user's mood, personal dietary preferences, or nutritional goals**, leading to generic and uninspired suggestions. This gap between emotional state and food recommendation results in a lack of personalization and user engagement.
- Additionally, cooking instructions are often static and require constant manual interaction, which can be inconvenient and frustrating—especially for beginners or individuals with accessibility needs.

- Hence, there is a need for an **intelligent, mood-aware system** that not only **recommends recipes** tailored to how a user feels but also provides **interactive, voice-guided cooking assistance**. The goal is to make cooking more intuitive, enjoyable, and aligned with the user's emotional and dietary context through the integration of **AI technologies** like **facial emotion recognition** and **voice-based interaction**.

1.2 OBJECTIVE OF PROJECT

The primary objective of this project is to develop an AI-powered web application, **MoodChef: Mood-Based Smart Recipe Recommender With Cooking Assistant**, that intelligently recommends recipes and guides users through cooking based on their current emotional state and dietary preferences.

The specific objectives include:

- To **detect the user's mood** using facial expression recognition via the webcam.
- To allow users to **select dietary preferences** (vegan, vegetarian, or non-vegetarian).
- To **recommend recipes** that align with both the user's mood and food preferences.
- To provide **hands-free cooking assistance** using a **voice-enabled interface**.
- To create an **interactive, user-friendly UI** that supports seamless navigation across modules.
- To demonstrate the application of **AI techniques** like emotion recognition and NLP in real-life tasks.

The project ultimately aims to make cooking a more **personalized, emotionally resonant**, and **user-friendly experience** by leveraging emerging technologies.

1.2 SCOPE

This project focuses on developing a web-based application that recommends recipes based on the user's mood and dietary preferences. It integrates AI-based facial emotion detection and voice-assisted cooking guidance. The system aims to enhance user engagement by delivering a personalized and interactive cooking experience.

Objective Points of the Scope:

1. To enable emotion-aware meal recommendations for elderly users based on real-time mood detection.
2. To assist users with cognitive or physical limitations through a hands-free, voice-guided cooking process.

3. To provide a personalized cooking experience that aligns with individual health goals and dietary restrictions.
4. To promote mental well-being and independence in elderly care through emotionally intelligent systems.
5. To develop a modular architecture that can integrate with future smart home or healthcare systems.

1.4 MOTIVATION

In today's busy and emotionally demanding world, food is not just a necessity—it is deeply connected to how we feel. Many people, especially students and working professionals, often find it difficult to decide what to cook or eat, especially when they are stressed, sad, or tired. Traditional recipe platforms offer a one-size-fits-all solution that ignores the emotional and mental state of the user.

The idea behind **MoodChef: Mood-Based Smart Recipe Recommender With Cooking Assistant** was born from the need to make cooking a more **emotionally aware**, **interactive**, and **enjoyable** experience. By combining **AI-powered mood detection** with smart recipe recommendations and **voice-guided cooking**, we aim to reduce decision fatigue, support mental well-being, and encourage healthier food choices. Our motivation is to use technology to transform everyday cooking into a personalized and uplifting activity that adapts to how the user feels.

CHAPTER 2

LITERATURE SURVEY

1. LITERATURE SURVEY

2.1 INTRODUCTION

With the growing intersection of artificial intelligence and daily life, emotion-aware systems have gained significant attention in domains such as healthcare, education, and human-computer interaction. In the context of food and nutrition, users often seek personalized solutions that cater not only to their dietary needs but also to their emotional states.

Recent advancements in facial emotion recognition, natural language processing, and voice interaction have enabled the development of intelligent systems capable of understanding and responding to human behavior. Several studies and applications have explored the use of AI-driven recommendation engines, but few have focused on the emotional aspect of user engagement, especially in cooking.

This literature review explores existing technologies and methodologies in mood detection, AI-based recipe recommendation systems, and voice-assisted applications. It provides the foundation upon which the MoodChef: Mood-Based Smart Recipe Recommender With Cooking Assistant system is built, identifying current gaps and highlighting the need for a more emotionally adaptive and user-friendly cooking assistant.

2.2 RELATED WORKS

Several research efforts and applications have explored the integration of artificial intelligence into personalized recommendation systems, facial emotion recognition, and voice-assisted interaction:

- **Facial Emotion Recognition:** Techniques like those used in Face-API.js and OpenCV have been widely adopted in affective computing. Studies have shown the effectiveness of facial landmarks and deep learning models in accurately identifying emotions such as happiness, sadness, anger, and surprise, which can be leveraged to tailor user experiences.
- **AI-Based Recipe Recommendation:** Traditional food recommendation systems often rely on collaborative filtering or content-based filtering. However, recent advancements have focused on contextual and health-aware models that factor in nutritional requirements, allergies, and user preferences. Projects like Yummly and IBM Chef Watson have demonstrated the potential of AI in culinary personalization.

- **Voice Assistants in Cooking:** Devices like Amazon Alexa and Google Assistant have been applied to provide voice-based step-by-step cooking instructions. Research has found that voice guidance improves usability and safety, especially in hands-busy environments like kitchens.
- **Mood-Centric Applications:** While mood-based recommendations have been applied in music and entertainment platforms (e.g., Spotify mood playlists), their use in food technology is relatively limited. This presents a novel opportunity for combining emotional AI with nutrition and cooking.

2.3 CHALLENGES ADDRESSED BY THE PROJECT

The MoodChef: Mood-Based Smart Recipe Recommender With Cooking Assistant system addresses several real-world challenges by integrating AI technologies into a personalized cooking platform:

1. Hands-Free Cooking Assistance

Users, especially beginners or elderly individuals, face difficulty following instructions while cooking. MoodChef: Mood-Based Smart Recipe Recommender With Cooking Assistant is a voice-enabled assistant that provides step-by-step cooking support without manual interaction.

2. Lack of Emotional Intelligence in Current Systems

Most food recommendation platforms are static and lack adaptive behavior. MoodChef: Mood-Based Smart Recipe Recommender With Cooking Assistance brings emotional intelligence into cooking, making the experience more human-centric and engaging.

3. Integration of Multiple AI Technologies

Combining facial emotion recognition, natural language processing, and voice synthesis presents a complex integration challenge, which MoodChef: Mood-Based Smart Recipe Recommender With Cooking Assistance successfully manages in a unified platform.

2.4 PROPOSED SYSTEM OVERVIEW

The proposed system, MoodChef: Mood-Based Smart Recipe Recommender With Cooking Assistance, is a smart web application that detects the user's mood through facial expression analysis and recommends recipes accordingly. Users can select dietary preferences, and the system filters recipes using AI-based matching. A voice-enabled assistant guides users through step-by-step cooking instructions, making the process intuitive and hands-free. The system combines emotion recognition, smart filtering, and voice interaction to deliver a personalized and engaging cooking experience.

2.5 TECHNOLOGICAL FRAMEWORK

The MoodChef: **Mood-Based Smart Recipe Recommender With Cooking Assistant** is built using:

- **Frontend:** HTML, CSS, JavaScript, Bootstrap for responsive UI; Face-API.js for mood detection.
- **Backend:** Node.js or Flask for server logic and integration.
- **AI Components:** Facial emotion recognition and smart filtering for recipe recommendations.
- **Voice Assistant:** Speech Synthesis API for hands-free cooking guidance.

2.6 USER BENEFITS

- Personalized recipe suggestions based on mood
- Saves time with automatic meal recommendations
- Hands-free voice-guided cooking
- Improves cooking skills, ideal for beginners
- Encourages mood-aware healthy eating
- Adds fun and innovation to cooking

CHAPTER 3

EXISTING SYSTEM

3. EXISTING SYSTEM

The concept of emotion-based recipe recommendations is relatively new, and most existing systems do not incorporate user mood into their logic. Below is an analysis of current systems relevant to the *MoodChef: Mood-Based Smart Recipe Recommender with Cooking Assistance* project:

1. Traditional Recipe Recommendation Apps

Description:

These systems allow users to browse or search for recipes based on parameters like ingredients, meal type, or cooking time (e.g., Tasty, Yummly, AllRecipes).

Limitations:

1. Do not personalize based on the user's emotional state
2. Require manual input and selection
3. Provide generic suggestions without user-specific context

2. Virtual Assistants and Cooking Guides

Description:

Voice assistants like Alexa, Google Assistant, and apps like SideChef offer voice-guided recipes and step-by-step instructions.

Limitations:

1. Do not detect or adapt to user mood
2. Only respond to explicit commands
3. No personalization or dynamic adaptation based on emotional input

3. Emotion Recognition Tools (Stand-alone)

Description:

Some research tools and apps can recognize emotions using facial expressions or voice tone but are not integrated into recipe or cooking systems.

Limitations:

1. Function independently without combining with recipe databases

2. Require technical know-how to use effectively
3. Not designed for end-to-end cooking assistance

3.1 DRAWBACKS OF EXSISTING SYSYEM

1. Lack of Personalization

Most existing systems provide generic recipe suggestions without considering the user's current emotional or psychological state.

2. Manual Effort Required

Traditional recipe platforms and food trackers require users to input preferences or search manually, which can be time-consuming and inconvenient.

3. No Emotion Detection Integration

Cooking apps and assistants do not integrate facial emotion recognition to tailor recommendations, limiting their interactivity and relevance.

4. Poor User Engagement

Due to the absence of emotional responsiveness or dynamic assistance, current systems fail to fully engage the user.

5. Limited Accessibility

Many solutions are not voice-enabled or hands-free, making them difficult to use during the cooking process, especially for users with disabilities.

CHAPTER 4

PROPOSED SYSTEM

4.PROPOSED SYSTEM

The proposed system is a **MoodChef: Mood-Based Smart Recipe Recommender with Cooking Assistance**, which combines **facial emotion recognition**, **AI-based personalized recipe suggestions**, and **voice-guided cooking support** into a single platform. Using a webcam, the system captures and analyzes the user's facial expressions to detect their emotional state (e.g., happy, sad, angry, neutral). Based on the detected mood, it suggests a suitable recipe to either complement or improve the user's emotional well-being. Once a recipe is selected, the system provides **step-by-step instructions** via **voice commands**, allowing for a **hands-free cooking experience**. This ensures greater user convenience, especially while cooking.

This integrated approach makes cooking not only easier and more interactive, but also emotionally intelligent and adaptive.

Merits of the Proposed System

1. Emotion-Based Personalization

Suggests recipes aligned with the user's current mood, offering comfort or uplifting food options.

2. Hands-Free Operation

Voice-enabled guidance allows users to follow recipes without needing to touch devices while cooking.

3. Time Efficiency

Instantly provides recipe suggestions without manual browsing or selection, saving time and effort.

4. Stress and Mood Management

Helps users manage stress or boost mood through smart meal suggestions.

5. Learning Support for Beginners

Provides easy-to-follow instructions, making it ideal for new or inexperienced cooks.

6. Future Scalability

Can be enhanced with dietary filters, regional preferences, and integration with smart kitchen devices.

CHAPTER 5

SOFTWARE AND HARDWARE REQUIREMENTS

5.1 SOFTWARE REQUIREMENTS

The software environment consists of development tools, libraries, and frameworks that support facial emotion detection, recipe recommendation logic, and voice assistance. The software requirements include:

- **Operating System:** The system is compatible with both Windows 10 and Linux (preferably Ubuntu 20.04 or higher).
- **Programming Language:** Python 3.x is used as the core development language due to its vast library support and simplicity.
- **Front-End Technologies:** HTML, CSS, and JavaScript are used for designing the user interface of the web-based application.
- **Back-End Framework:** Flask is used as the backend Python web framework to integrate the front end with the core logic and APIs.
- **Required Libraries:** Key Python libraries needed are OpenCV for image processing, TensorFlow/Keras for emotion detection, NumPy and pandas for data handling, pyttsx3 for text-to-speech, and speech_recognition for voice input.

5.2 HARDWARE REQUIREMENTS

The hardware setup includes a basic computing system and peripheral devices necessary for facial emotion recognition and voice-based interactions. The minimum hardware requirements are:

- **Processor:** Intel i3 or higher is recommended to handle video processing and application logic smoothly.
- **RAM:** At least 4 GB of RAM is necessary to manage the execution of the system without lag.
- **Hard Disk:** A minimum of 100 GB of available disk space is needed to store libraries, project files, and datasets.
- **Webcam:** A built-in or external webcam is required for capturing facial expressions and performing mood detection.
- **Microphone and Speaker:** These are essential for enabling voice interaction between the system and the user.

CHAPTER 6

SYSTEM DESIGN

6.1 SOFTWARE DESIGN

The cooking assistant application integrates various modules, including mood-based recipe recommendations, authentication, navigation, and voice-enabled step-by-step guidance, to enhance the user experience and provide seamless interaction.

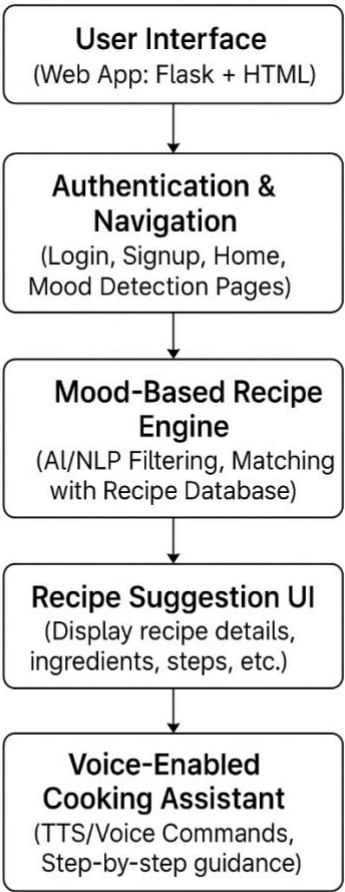
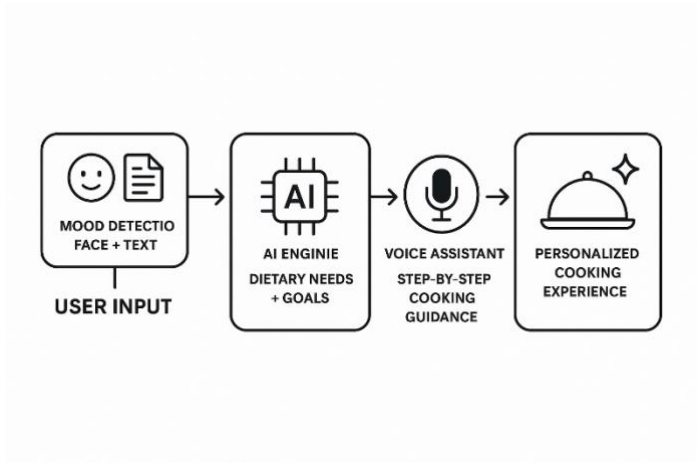


Fig 6.1 Architecture Diagram

6.2 UML DIAGRAMS

1. Use Case Diagram

This **use case diagram** visually represents the interactions between a user and a cooking assistant system. It highlights various functionalities the system offers, such as **mood detection via webcam**, **recipe recommendations**, **voice-guided cooking**, and **saving favorite recipes**.

The diagram includes:

- **Actors:** The User, Emotion Detection System, Recipe Engine, and Voice Assistant.
- **Use Cases:** Actions like **Login/Register**, **Detect Mood via Webcam**, **Set Preferences**, **Recommend Recipes**, **Start Cooking Mode**, and **Voice Guidance**.
- **Interactions:** Each component interacts with different use cases to create a smooth and intuitive cooking experience.

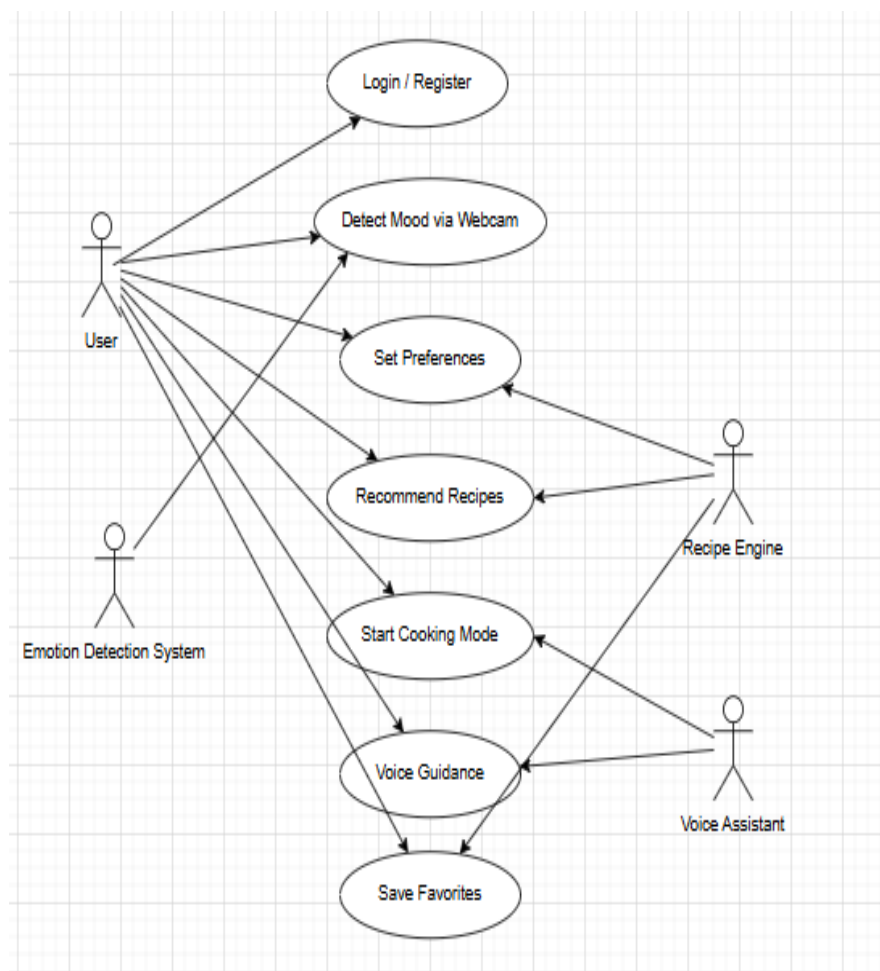


Fig 6.2.1 Use Case Diagram

2. Class Diagram

This class diagram visually represents the interactions between a user and a cooking assistant system. It highlights various functionalities the system offers, such as mood detection via webcam, recipe recommendations, voice-guided cooking, and saving favorite recipes.

- Actors: **The User, Emotion Detection System, Recipe Engine, and Voice Assistant.**
- Use Cases: **Actions like Login/Register, Detect Mood, Set Preferences, Recommend Recipes, Start Cooking Mode, and Voice Guidance.**
- Interactions: **Each component interacts with different use cases to create a smooth and intuitive cooking experience.**

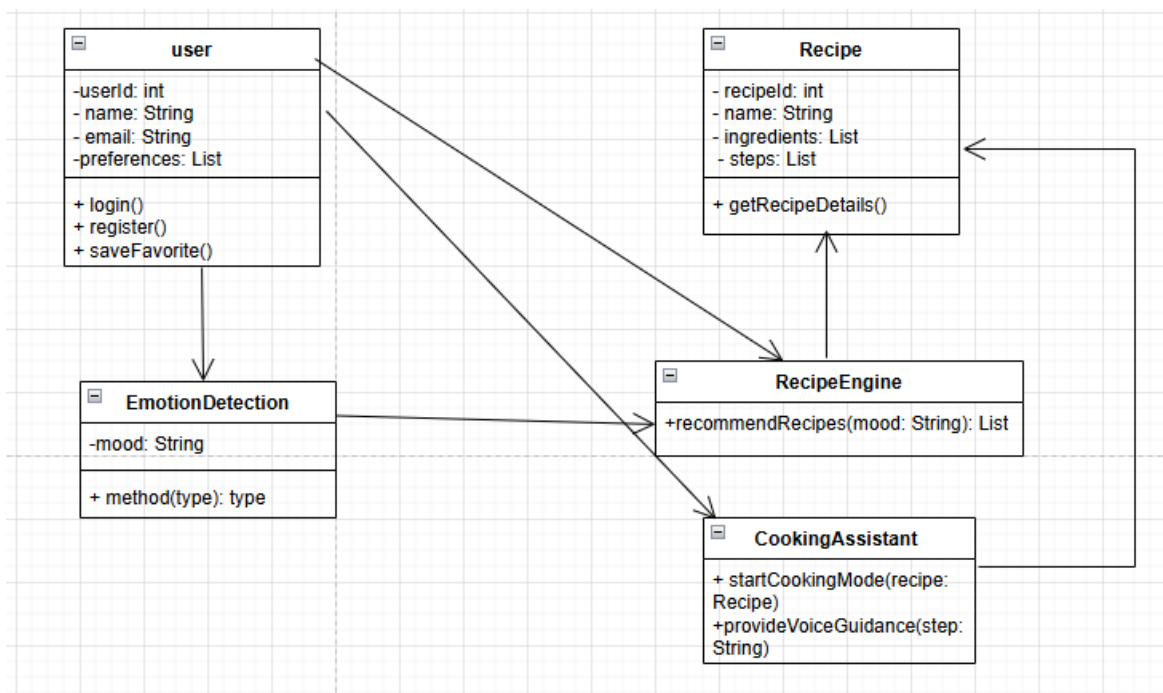


Fig 6.2.3 Class Diagram

3. Sequence Diagram

This sequence diagram visually represents the interactions between a user and a cooking assistant system. It highlights various functionalities the system offers, such as mood detection via webcam, recipe recommendations, voice-guided cooking, and saving favorite recipes.

- *- Actors: **The User, Emotion Detection System, Recipe Engine, and Voice Assistant.**
- *- Use Cases: **Actions like Login/Register, Detect Mood, Set Preferences, Recommend Recipes, Start Cooking Mode, and Voice Guidance.**
- *- Interactions: **Each component interacts with different use cases to create a smooth and intuitive cooking experience.**

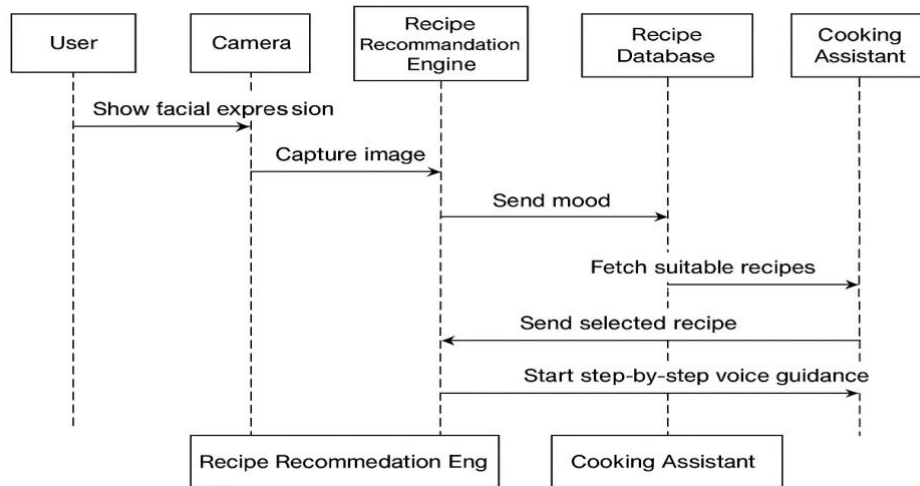


Fig 6.2.3 Sequence Diagram

4. Activity Diagram

This activity case diagram visually represents the interactions between a user and a cooking assistant system. It highlights various functionalities the system offers, such as **mood detection via webcam, recipe recommendations, voice-guided cooking, and saving favorite recipes.**

- *- Actors: **The User, Emotion Detection System, Recipe Engine, and Voice Assistant.**
- *- Use Cases: **Actions like Login/Register, Detect Mood, Set Preferences, Recommend Recipes, Start Cooking Mode, and Voice Guidance.**
- *- Interactions: **Each component interacts with different use cases to create a smooth and intuitive cooking experience.**

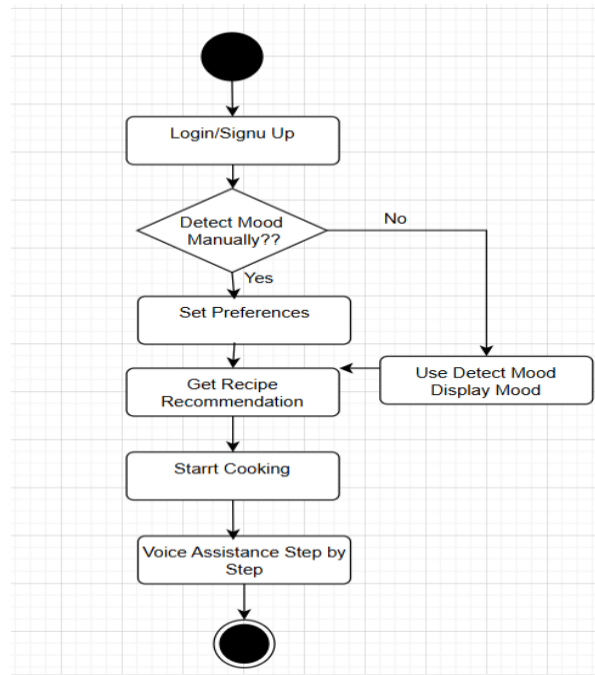


Fig 6.2.3 Activity Diagram

6.3 MODULES

The proposed system is structured into several functional modules to ensure modularity, clarity, and ease of implementation. Each module handles a specific task within the overall application. The major modules are described below:

1. Facial Emotion Detection Module

This module is responsible for capturing the user's facial expression using the webcam and analyzing it to detect their current emotional state. It uses machine learning models trained on facial landmarks and features to classify emotions such as happy, sad, angry, neutral, etc.

- **Tools Used:** OpenCV, TensorFlow/Keras
- **Functionality:** Real-time face detection and emotion classification

2. Recipe Recommendation Module

Based on the emotion identified in the previous module, this module selects appropriate recipes. For example, comforting foods may be suggested if the user appears sad, while refreshing recipes may be shown if the user is angry or stressed.

- **Logic:** Emotion-to-recipe mapping
- **Data:** Predefined recipe dataset categorized by emotional relevance

3. Voice Assistant Module

This module provides step-by-step cooking instructions using a voice interface. It enhances the user experience by enabling hands-free interaction, which is particularly useful while cooking.

- **Tools Used:** pyttsx3 (text-to-speech), speech_recognition
- **Features:** Voice prompts, repeat steps on request

4. Web Interface Module

A user-friendly interface built using HTML, CSS, and JavaScript allows users to interact with the system, view suggested recipes, and control the assistant. This module serves as the front end of the application.

- **Design:** Clean layout with image-based recipe display
- **Functionality:** Input display, navigation controls, recipe viewer

5. Backend Integration Module

This module acts as a bridge between the frontend and the processing logic. It handles the emotion data, recipe suggestion logic, and communication with the voice assistant.

- **Framework Used:** Flask
- **Responsibilities:** Routing, API handling, session management

6. Dataset Management Module

This module manages the recipe database and any training datasets used for emotion recognition. It allows the system to be updated with new recipes or retrained with improved datasets.

- **Content:** Recipe categories, facial expression datasets
- **Scalability:** Allows addition of dietary filters and region-based cuisines

Technical Tools and Packages

- Python 3.x
- OpenCV
- TensorFlow / Keras
- Flask
- HTML/CSS/JavaScript
- speech_recognition
- pyttsx3
- NumPy
- pandas
- Jupyter Notebook / Visual Studio Code (IDE)

CHAPTER 7

IMPLEMENTATION

7. IMPLEMENTATION

1. Project Architecture

1. Frontend:

- Build a user-friendly web interface using **HTML, CSS, and JavaScript**.
- Implement **webcam access** through JavaScript to capture user facial expressions.
- Display emotion-based recipe suggestions and step-by-step cooking instructions.
- Add interactive buttons for voice navigation (e.g., "Start", "Next Step", "Repeat").

2. Backend:

- Use **Python (Flask)** as the backend server to:
 - Process webcam input and run facial emotion recognition models.
 - Manage recipe data and map detected moods to relevant food suggestions.
 - Integrate voice command processing and speech synthesis.

3. Machine Learning:

- Train and deploy a **Convolutional Neural Network (CNN)** model using **TensorFlow/Keras** for emotion detection.
- Model detects expressions like *Happy, Sad, Angry, Neutral*, etc.

4. Voice Assistant:

- Integrate **speech recognition (speech_recognition)** to accept voice commands.
- Use **pyttsx3** to deliver real-time audio instructions for cooking steps.

2. Features Implementation

A. Emotion Detection

- Use **OpenCV** to capture facial input from webcam.
- Analyze facial expressions using a pre-trained CNN model.

B. Recipe Recommendation

- Map mood to specific food categories (e.g., Happy → Sweet, Sad → Spicy).
- Select suitable recipes from a pre-defined dataset using **pandas**.

C. Voice-Enabled Cooking Assistant

- Convert instructions into audio using **pyttsx3**.
- Accept user voice commands like “Next Step”, “Repeat Step”, or “Stop”.

D. User Interaction & Feedback

- Allow users to rate or like recipes for future personalization.
- Store previous sessions and preferences (optional feature extension).

3. Tools and Technologies

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python, Flask
- **Machine Learning:** TensorFlow/Keras, OpenCV
- **Voice:** pyttsx3 (Text-to-Speech), speech_recognition (Speech-to-Text)
- **Data Processing:** pandas, NumPy
- **IDE:** Visual Studio Code / Jupyter Notebook

4. Testing and Deployment

Testing:

- Validate emotion detection accuracy using different facial expressions.
- Verify recipe mapping logic for each detected emotion.
- Test voice assistant functionality across multiple steps.
- Ensure browser compatibility and camera access on multiple devices.

Deployment:

- Host using a local Flask server or deploy via services like **Render**, **Heroku**, or **Replit**.
- Ensure access to required Python dependencies and machine learning model files.

7.1 SAMPLE SOURCE CODE:

```
from flask import Flask, render_template, request, redirect, url_for, session

import sqlite3

from werkzeug.security import generate_password_hash, check_password_hash

app = Flask(__name__)

app.secret_key = 'your_secret_key' # Needed for session management!

# ----- ROUTES -----

@app.route('/')

def index():

    return redirect(url_for('login'))

@app.route('/signup', methods=['GET', 'POST'])

def signup():

    if request.method == 'POST':

        username = request.form['name']

        email = request.form['email']

        password = request.form['password']

        hashed_pw = generate_password_hash(password)

        # Add placeholders for other fields

        first_name = "

        last_name = "

        date_of_birth = "

        conn = sqlite3.connect('users.db')

        c = conn.cursor()

        try:
```

```

c.execute("""

INSERT INTO users

(username, email, password, first_name, last_name, date_of_birth)

VALUES (?, ?, ?, ?, ?, ?)

""", (username, email, hashed_pw, first_name, last_name, date_of_birth))

conn.commit()

except sqlite3.IntegrityError:

    conn.close()

    return "Email already registered. Please use a different email."

conn.close()

session['user'] = username

return redirect(url_for('login'))

return render_template('signup.html')

@app.route('/login', methods=['GET', 'POST'])

def login():

    if request.method == 'POST':

        email = request.form['email']

        password = request.form['password']

        conn = sqlite3.connect('users.db')

        conn.row_factory = sqlite3.Row

        c = conn.cursor()

        c.execute("SELECT * FROM users WHERE email = ?", (email,))

        user = c.fetchone()

        conn.close()

```

```

    if user and check_password_hash(user['password'], password):

        session['user'] = user['username']

        return redirect(url_for('home'))

    else:

        return "Login failed. Invalid email or password."

    return render_template('login.html')

@app.route('/home')
def home():

    if 'user' not in session:

        return redirect(url_for('login'))

    return render_template('home.html', username=session['user'])

@app.route('/mood')
def mood():

    if 'user' not in session:

        return redirect(url_for('login'))

    return render_template('mood.html')

@app.route('/detect')
def detect():

    if 'user' not in session:

        return redirect(url_for('login'))

    return render_template('detect.html')

# ----- MAIN -----

if __name__ == '__main__':

    app.run(debug=True)

```

Home Page

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8" />

<meta name="viewport" content="width=device-width, initial-scale=1.0" />

<title>Choose Mood - MoodChef</title>

<script src="https://cdn.tailwindcss.com"></script>

<link rel="preconnect" href="https://fonts.googleapis.com">

<link      href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;600&display=swap"
rel="stylesheet">

<style>

body {

font-family: 'Poppins', sans-serif;

}

.hidden {

display: none;

}

</style>

</head>

<body class="bg-gradient-to-tr from-pink-300 via-purple-400 to-indigo-500 min-h-screen flex flex-col
items-center justify-center text-white relative">

<!-- ♦ Quick Hub -->

<div class="absolute top-4 right-4 z-50 group">



<div id="quickHub" class="hidden mt-2 bg-white text-gray-800 rounded-lg shadow-lg w-64 absolute
right-0 p-4 space-y-2">

<h3 class="text-lg font-semibold text-indigo-600 mb-2">MoodChef Quick Hub</h3>

<button onclick="showChatbot()" class="w-full text-left hover:bg-gray-100 px-2 py-1 rounded"><img alt="chatbot icon" data-bbox="895 248 920 268"/>
Today's Mood Recipe</button>

<button onclick="location.href='/detect'" class="w-full text-left hover:bg-gray-100 px-2 py-1
rounded"><img alt="refresh icon" data-bbox="172 338 195 353"/> Re-detect Mood</button>

</div>

</div>

<!-- ♦ Mood Selection -->

<div class="p-10 rounded-3xl shadow-2xl text-center w-[90%] max-w-xl bg-white bg-opacity-10
backdrop-blur-lg">

<h1 class="text-3xl font-semibold mb-6">Choose Your Mood</h1>

<p class="text-lg text-gray-100 mb-4">Would you like to select your mood manually or use the mood
detected?</p>

<button onclick="useDetectedMood()" class="bg-yellow-400 hover:bg-yellow-300 text-gray-900 font-
semibold py-2 px-6 rounded-full transition duration-300 transform hover:scale-105 shadow-lg mb-4">

Use Detected Mood

</button>

<div>

<p class="text-gray-100 mb-4">Or, select a mood manually:</p>

<select id="manualMood" class="p-2 rounded-md text-gray-800">

<option value="happy">Happy</option>

<option value="sad">Sad</option>

```

```
<option value="angry">Angry</option>
```

```
<option value="surprised">Surprised</option>
```

```
</select>
```

```
<button onclick="submitMood()" class="bg-yellow-400 hover:bg-yellow-300 text-gray-900 font-  
semibold py-2 px-6 rounded-full transition duration-300 transform hover:scale-105 shadow-lg mt-4">
```

Submit Mood

```
</button>
```

```
</div>
```

```
<div id="dietaryPreferences" class="hidden mt-8">
```

```
<p class="text-lg text-gray-100 mb-4">Please select your dietary preferences:</p>
```

```
<select id="dietaryChoice" class="p-2 rounded-md text-gray-800">
```

```
<option value="vegan">Vegan</option>
```

```
<option value="vegetarian">Vegetarian</option>
```

```
<option value="non-vegetarian">Non-Vegetarian</option>
```

```
</select>....
```

CHAPTER 8

RESULTS

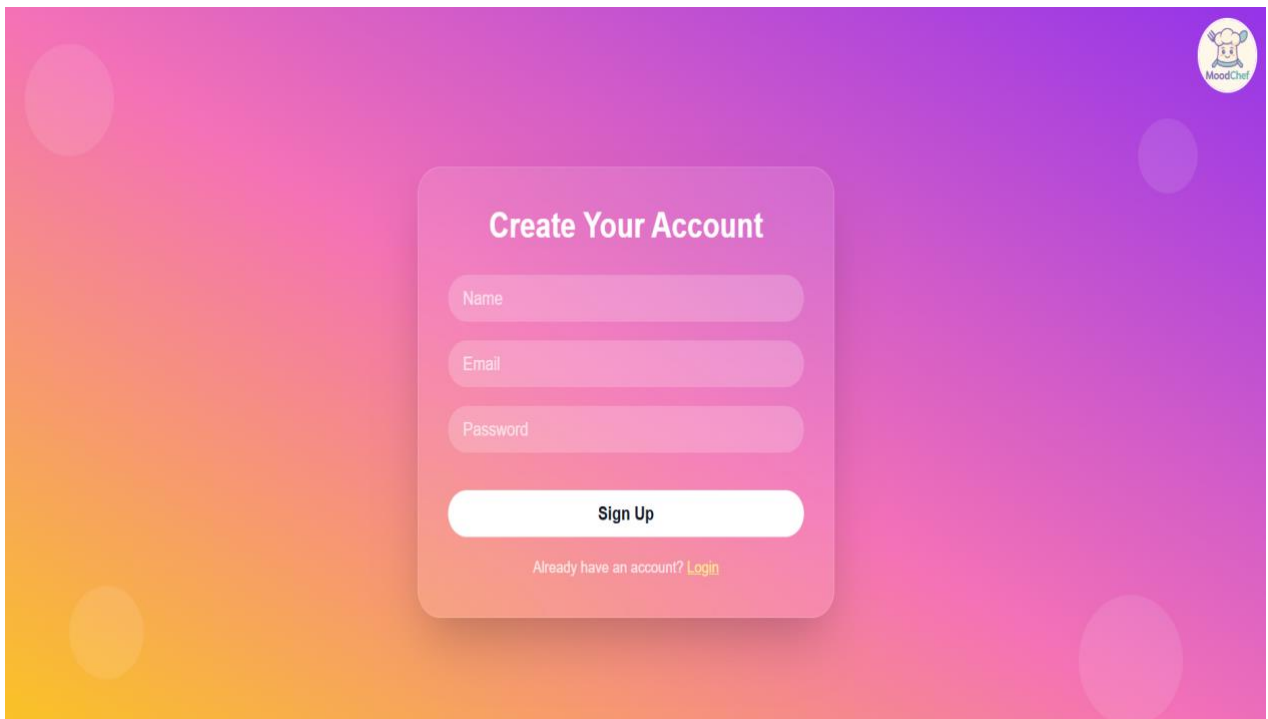


Fig 8.1 Output Snapshot 1

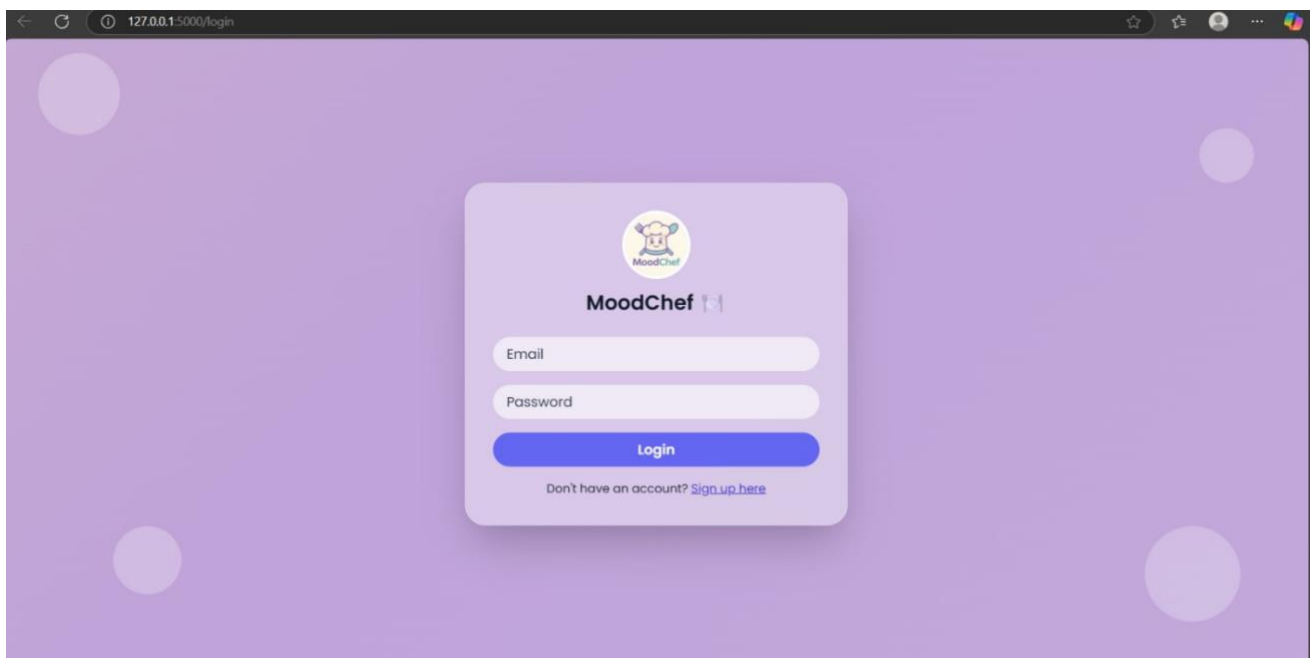


Fig 8.2 Output Snapshot 2

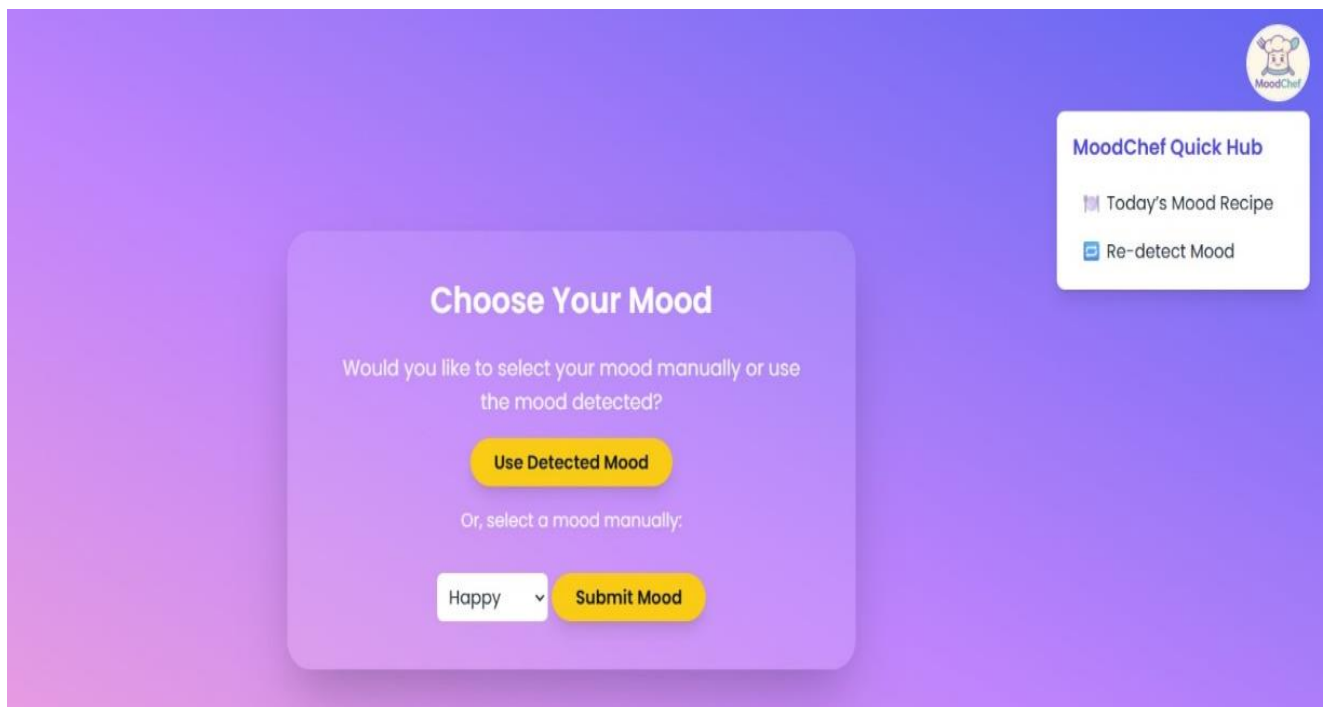


Fig 8.3 Output Snapshot 3

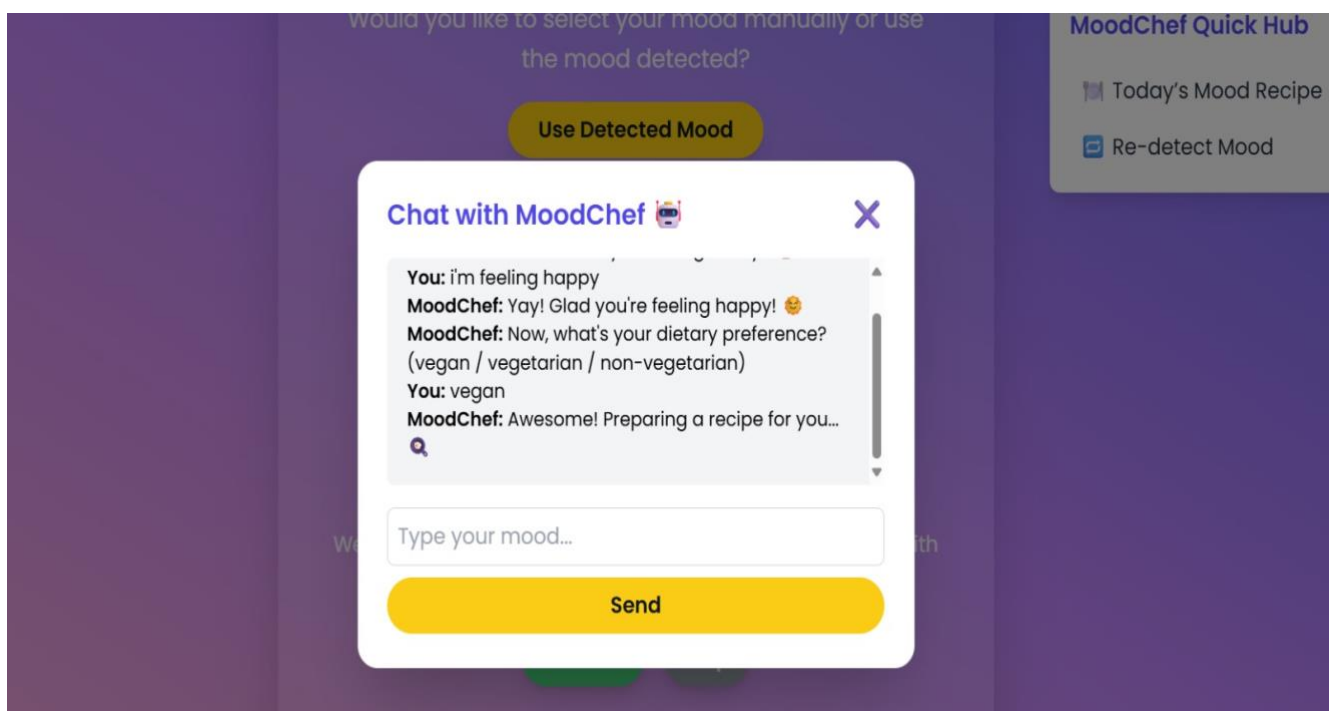


Fig 8.4 Output Snapshot 4



Choose Your Mood

Would you like to select your mood manually or use the mood detected?

Use Detected Mood

Or, select a mood manually:

Happy **Submit Mood**

Please select your dietary preferences:

Vegan **Submit Dietary Preferences**

Select Nutrition Goal:

Balanced

Fig 8.5 Output Snapshot 5



Choose Your Mood

Would you like to select your mood manually or use the mood detected?

Use Detected Mood

Or, select a mood manually:

Happy **Submit Mood**

Your Suggested Recipe

Fruity Vegan Smoothie

Ingredients: Berries, Bananas, Oranges, Chia seeds, Maple syrup, Almond milk, Ice, Mint leaves

Step 1: In a blender, combine berries, bananas, oranges, and chia seeds.

Step 2: Add a splash of maple syrup and almond milk to the blender.

Step 3: Add a handful of ice to make the smoothie cold and refreshing.

Step 4: Blend until smooth and creamy.

Fig 8.6 Output Snapshot 6

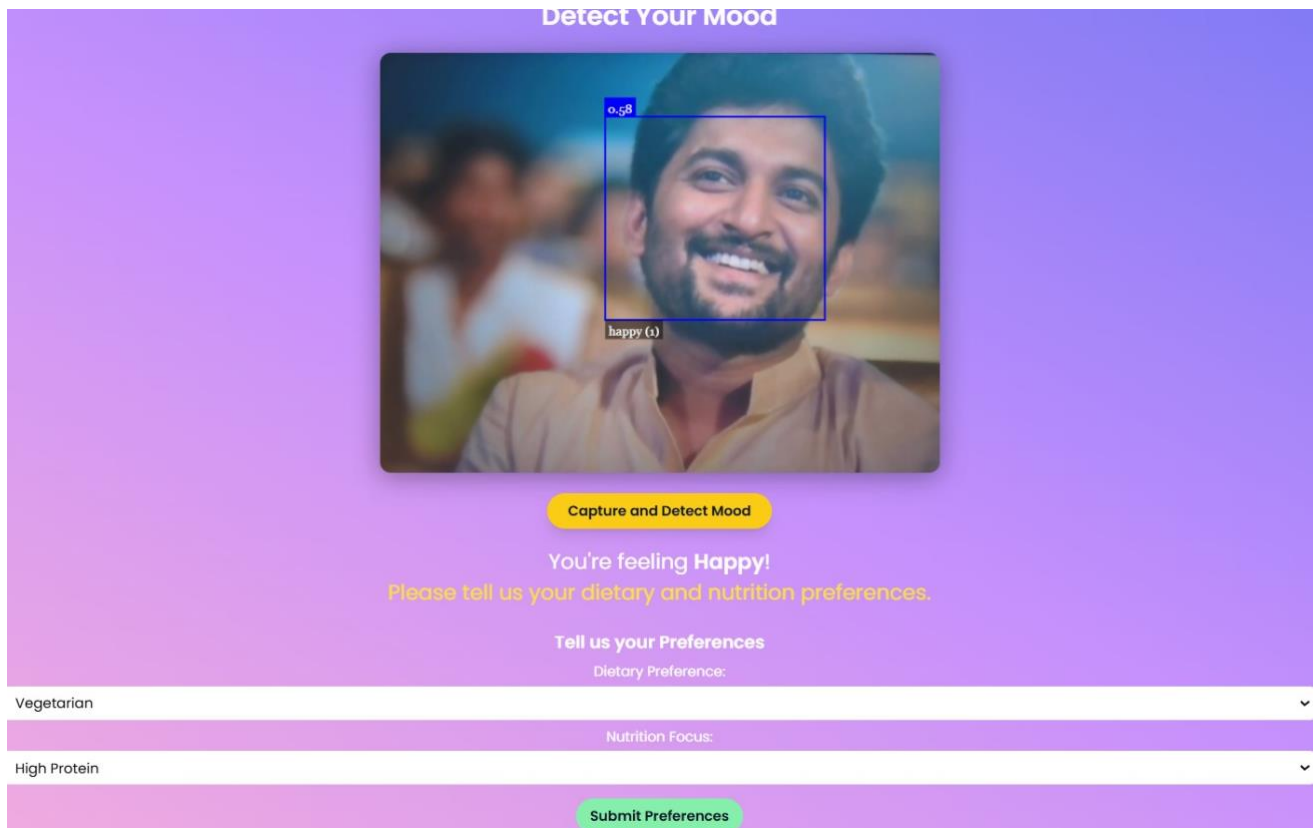


Fig 8.1 Output Snapshot 7



Fig 8.7 Output Snapshot 8

CHAPTER 9

CONCLUSION

7. CONCLUSION

The "Mood-Based Smart Recipe Recommender with Cooking Assistance" project presents an innovative and user-centric approach to enhancing the cooking experience by integrating emotion recognition, intelligent recommendation systems, and voice-enabled guidance. By detecting a user's mood through facial expressions and suggesting personalized recipes accordingly, the system ensures emotional well-being and convenience in daily cooking tasks.

The integration of technologies such as computer vision, machine learning, and natural language processing allows for a seamless interaction between the user and the system. The voice-assisted cooking module adds hands-free utility, making the platform especially helpful in real-time kitchen scenarios. This intelligent assistant not only saves time and effort but also encourages emotional engagement with the cooking process.

Overall, this project effectively bridges the gap between artificial intelligence and day-to-day lifestyle, offering a practical and interactive solution that could be extended in future to support features like dietary preferences, regional cuisine recommendations, and integration with smart kitchen appliances.

CHAPTER 10

FUTURE SCOPE

10.FUTURE WORK

The proposed system holds significant potential for expansion and real-world applicability. Future enhancements can greatly improve its functionality, personalization, and accessibility. Some key future scopes include:

1. Diet-Based Recommendations
 - Integrate dietary preferences (e.g., vegan, diabetic, low-carb) to provide health-specific recipe suggestions.
2. Multilingual Support
 - Enable the voice assistant to understand and respond in multiple regional languages for broader user accessibility.
3. User Profile and History Tracking
 - Maintain user history, mood logs, and previously recommended recipes to improve personalization over time.
4. Integration with Smart Kitchen Devices
 - Connect with IoT-enabled kitchen appliances (e.g., smart ovens, fridges) for automated cooking assistance and inventory tracking.
5. Mobile App Version
 - Develop a cross-platform mobile application for more flexible and on-the-go usage.
6. Nutritional Analysis
 - Provide nutritional value breakdown of each recipe to promote healthier eating habits.
7. Enhanced Emotion Detection
 - Improve the emotion recognition model by incorporating advanced facial expression datasets and even voice tone analysis for higher accuracy.
8. Cloud-Based Model Hosting
 - Deploy machine learning models on cloud platforms for scalability and faster processing.

REFERENCES

1. *A Cloud-Based Smart Expiry System Using QR Code*, IEEE, 2018. [Online]. Available: <https://ieeexplore.ieee.org/document/8500140>
2. *FruitfulInsight: IoT and ML-Based System to Predict Expiry of Fruits*, IEEE, 2024. [Online]. Available: <https://ieeexplore.ieee.org/document/10533165>
3. *Internet of Things Based Smart Refrigeration System*, IEEE, 2023. [Online]. Available: <https://ieeexplore.ieee.org/document/10295021>
4. *Sensors for Smart Packaging in Healthcare and Food Industry*, IEEE, 2021. [Online]. Available: <https://ieeexplore.ieee.org/document/9639598>
5. *International Research Journal of Engineering and Technology (IRJET)*. [Online]. Available: <https://www.irjet.net>
6. *International Research Journal of Modernization in Engineering Technology and Science (IRJMETS)*. [Online]. Available: <https://www.irjmets.com>
7. *International Journal of Innovative Technology and Exploring Engineering (IJITEE)*. [Online]. Available: <https://www.ijitee.org>
8. *Mood-Based Smart Recipe Recommender with Cooking Assistance*, Unpublished Project, 2025.

WEBSITES REFERENCES

- IRJET - <https://www.irjet.net>
- IRJMETS - <https://www.irjmets.com>
- IJITEE - <https://www.ijitee.org>